Laxmi Lamichhane

Full-Stack Engineer · Pokhara, Nepal · lc.llaxmi@gmail.com · laxmilamichhane.com.np · github.com/llaxmi

Full-stack engineer shipping production TypeScript across the stack — NestJS services, Postgres/Prisma data models and REST integrations on the backend, React and Next.js interfaces on the front. Currently building AI-backed data enrichment features for a SaaS platform, with a research background in retrieval pipelines and vector search.

EXPERIENCE

Full-Stack Developer — SyncGTM

Feb 2025 – Present

Sydney, NSW (Remote)

- Own features end to end on a Next.js + NestJS TypeScript SaaS platform — designing the API endpoints and data models behind a feature as well as the interface that consumes them.
- Designed and implemented a **Sync Agent** that drives automated data enrichment through structured LLM output, including schema-validated responses and error handling for unreliable model results.
- Built an **MCP Server integration** exposing platform capabilities as tools to external AI clients.
- Built backend integration points against third-party APIs, handling authentication, pagination, rate limits and failure retries so enrichment jobs degrade gracefully rather than drop data.
- Architected the client data layer with TanStack Query — organised query-key management, optimistic updates and a global error-handling system — and built advanced TanStack Table views over large datasets.
- Work directly with designers and backend engineers in a fast-paced startup to scope, build and ship product features end to end.

Junior Developer — Codse Tech Pvt. Ltd.

Jan 2024 – Sep 2024

Darbarthok, Pokhara

- Built and maintained React.js + TypeScript interfaces wired to REST APIs, applying state management, hooks and lifecycle patterns.
- Enforced type safety across the codebase with TypeScript, reducing runtime errors and review churn.
- Collaborated through Git/GitHub workflows — branching, pull requests and code review — and contributed features and bug fixes to the **Animata** open-source project.

PROJECTS

Full list at laxmilamichhane.com.np

ToS-RAG — Final Year Research Project

Aug 2026

A controlled experiment on retrieval-augmented generation pipeline design over legal Terms of Service documents, shipped with a public question-answering demo.

- Built the full backend pipeline — document ingestion, chunking, embedding and vector search over **PostgreSQL + pgvector**, served through a **Hono** API with Prisma-modelled schemas in a Turborepo monorepo.
- Designed and ran a 15-configuration sweep (5 chunking strategies × 3 chunk sizes) against 20 ground-truth questions with the generator held fixed, producing 300 fully traceable runs; a second phase benchmarked a self-hosted Llama 3.1 8B against Claude Opus under the winning configuration.
- Hand-rolled five chunking strategies preserving exact character offsets into the source document, enabling character-span retrieval precision/recall (LegalBench-RAG style) alongside CRAG-style LLM-judge scoring and SQuAD F1/EM.
- Shipped the results as a React dashboard and an interactive ask interface where every answer cites its source clause.

Stack: TypeScript, Hono, Prisma, PostgreSQL + pgvector, Python (SciPy, statsmodels), Ollama / Llama 3.1 8B, Claude API, React 19, Vite, Turborepo

GENA — AI Quiz Generation Platform

Nov 2025

A full-stack web application that turns any study material into personalised quizzes.

- Designed the relational schema and data access layer with **Prisma and PostgreSQL** (users, materials, generated quizzes and attempt history).
- Built the server-side generation flow on **Next.js route handlers** — file upload and parsing, prompting a self-hosted **Ollama** model, and validating model output into a strict quiz structure before persisting it.

- Containerised the app and its Postgres and Ollama dependencies with **Docker Compose** for reproducible local and deployment environments.
- Built the responsive front end in Next.js and Tailwind CSS.

Stack: Next.js, TypeScript, Prisma, PostgreSQL, Ollama, Docker, Tailwind CSS

IdeaPulse — Startup Idea Similarity Search

Oct 2024

A tool that checks whether a startup idea similar to yours has already been funded by Y Combinator.

- Built the ingestion and retrieval pipeline — embedding a corpus of YC-funded companies into a **vector store** and serving semantic similarity search over it with **LangChain**.
- Ran generation locally through Ollama models to keep results context-aware without external API costs.
- Built the React + Vite interface for querying and exploring matches.

Stack: React, Vite, LangChain, Vector Store, Ollama

Wander Pokhara — Community Travel Guide

Aug 2024

- Built a server-rendered **Node.js / Express** application with RESTful routes and persisted listings, letting users discover and submit hidden places around Pokhara.

Stack: Node.js, Express, Tailwind CSS

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C, C++, SQL

Backend: NestJS, Node.js, Express, Hono, Next.js route handlers, REST API design, authentication, background jobs

Frontend: React, Next.js, React Native / Expo, TanStack Query, TanStack Table, Tailwind CSS

Databases & Data: PostgreSQL, MySQL, MongoDB, Supabase, Prisma, pgvector, vector stores

AI Engineering: RAG pipelines, LLM APIs (Claude, OpenAI), structured output, prompt engineering, LangChain, Ollama, MCP

DevOps & Tools: Docker, Git/GitHub, AWS, Vercel, Postman, Vite, Figma

EDUCATION

Gandaki College of Engineering and Science

Sep 2022 – Sep 2026

Bachelor in Software Engineering

Sagarmatha Secondary School

Sep 2020 – Apr 2022

Higher Secondary Education

ACHIEVEMENTS

- **First Prize, Provincial Ideathon** — Code For Change, Nov 2024: prototype detecting plant disease from leaf images.
- **Git & GitHub Mentor** — GCES Girls Tech, Dec 2024: taught collaborative version-control workflows and open-source contribution.
- **Codefest Hackathon 2024** — Code For Change; **Hour of Code** instructor — Code.org, Jan 2025.